HDOC-7

Question 1:

Write a C program to find torque when force and perpendicular distance are given.

Task 1: Understanding the problem statement:

- Our objective is to develop a C program that finds the torque when the force and its perpendicular distance are given
- Required input for the C program will be:
  - Force
  - Perpendicular distance
- Formula used in the program is as follows:
  - Torque = Force x perpendicular distance

Task 2: Standard test case table:

Test Case	Input(s) (Force, Distance)	Expected Output(s)	Actual Output(s)	Result (Pass/Fail)
1	10 N, 2 m	Torque = 20.00 N·m	Torque = 20.00 N·m	PASS
2	7.4 N, 5 m	Torque = 37.00 N·m	Torque = 37.00 N·m	PASS
3	0 N, 4 m	Torque = 00.00 N·m	Torque = 00.00 N·m	PASS
4	30 N, 0.5 m	Torque = 15.00 N·m	Torque = 15.00 N·m	PASS
5	8 N, 8 m	Torque = 36.00 N·m	Torque = 36.00 N·m	PASS

Task 3: The Algorithm:

- Start.
- Declare three floating-point variables:
  - force
  - distance
  - torque
- Read the values of force and perpendicular distance from the user.
- Calculate torque using the formula:
- **Torque = Force × Perpendicular Distance**
- Display the calculated torque.
- Stop.

Task 4: C program:

```
#include <stdio.h>
 int main()
 {
 float force, distance, torque;
printf("Enter force (in Newton): ");
 scanf("%f", &force);
 printf("Enter perpendicular distance (in meter): ");
 scanf("%f", &distance);
 torque = force * distance;
 printf("\nTorque = %.2f N.m\n", torque);
return 0;
}
```

Task 5: Test cases output:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter force (in Newton): 10
Enter perpendicular distance (in meter): 2

Torque = 20.00 N.m
```

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter force (in Newton): 7.4
Enter perpendicular distance (in meter): 5

Torque = 37.00 N.m
```

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter force (in Newton): 0
Enter perpendicular distance (in meter): 4

Torque = 0.00 N.m
```

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter force (in Newton): 30
Enter perpendicular distance (in meter): 0.5

Torque = 15.00 N.m
```

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter force (in Newton): 8
Enter perpendicular distance (in meter): 8

Torque = 64.00 N.m
```

Question 2:

Write a C program to find viscous force using Stoke's law when viscosity, radius, and velocity are given.

Task 1: Understanding the problem statement:

- Our object is to write a C program to calculate viscous force, when the coefficient of viscosity , radius and velocity are given.
- Input required:
    - Coefficient of viscosity in Pa·s (N·s/m²)
    - Radius in meters (m)
    - Velocity in meters/second (m/s)
- Output:
    - Viscous Force (F) in Newton (N)
- Formula:

$$F = 6\pi\eta rv$$

where:

- F= Viscous Force (N)
- $\pi = 3.14$
- η= Coefficient of viscosity
- r= Radius of the sphere
- v= Velocity of the sphere

Step 2: Standard Test Case Table:

Test Case	Input(s) (η, r, v)	Expected Output(s)	Actual Output(s)	Result (Pass/Fail)
1	0.5, 0.02, 3	0.57 N	0.57 N	PASS
2	1.0, 0.05, 2	1.88 N	1.88 N	PASS
3	0.8, 0.01, 5	0.75 N	0.75 N	PASS
4	0.6, 0.03, 0	0.00 N	0.00 N	PASS
5	1.2, 0.04, 1.5	1.36 N	1.36 N	PASS

Task 3: The Algorithm:

- Start.
- Declare four floating-point variables:
  - viscosity
  - radius
  - velocity
  - force
- Define the constant value of π as **3.14.**
- Read the values of viscosity, radius, and velocity from the user.
- Calculate the viscous force using Stoke's Law:
  Force = 6 × π × Viscosity × Radius × Velocity
- Display the calculated viscous force.
- Stop.

Task 4: C program:

```c
#include <stdio.h>
#define PI 3.14159
int main()
 {
     float viscosity, radius, velocity, force;
     printf("Enter coefficient of viscosity (Pa.s): ");
     scanf("%f", &viscosity);

    printf("Enter radius of the sphere (m): ");
    scanf("%f", &radius);

    printf("Enter velocity (m/s): ");
    scanf("%f", &velocity);
    force = 6 * PI * viscosity * radius * velocity;

    printf("\nViscous Force = %.2f N\n", force);
    return 0;
}
```

Task 5: Test cases output:

Test case1:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter coefficient of viscosity (Pa.s): 0.5
Enter radius of the sphere (m): 0.02
Enter velocity (m/s): 3

Viscous force = 0.57 N
```

Test case2:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter coefficient of viscosity (Pa.s): 1
Enter radius of the sphere (m): 0.05
Enter velocity (m/s): 2

Viscous force = 1.88 N
```

Test case3:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter coefficient of viscosity (Pa.s): 0.8
Enter radius of the sphere (m): 0.01
Enter velocity (m/s): 5

Viscous force = 0.75 N
```

Test case4:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter coefficient of viscosity (Pa.s): 0.6
Enter radius of the sphere (m): 0.03
Enter velocity (m/s): 0

Viscous force = 0.00 N
```

Test case5:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter coefficient of viscosity (Pa.s): 1.2
Enter radius of the sphere (m): 0.04
Enter velocity (m/s): 1.5

Viscous force = 1.36 N
```

Question 3:

Write a C program to find speed of light in a medium when refractive index is given.

Task 1: Understanding the problem statement:

- The objective is to write a C program to calculate the speed of light in a medium when the refractive index of the medium is given.
- Input
  - Refractive Index (n)
- Output
  - Speed of Light in the Medium (v) in meters per second (m/s)
- Formula

The speed of light in a medium is given by:

$$\boxed{v = \frac{c}{n}}$$

where:

- v = Speed of light in the medium (m/s)
- c = Speed of light in vacuum = 3×10^8 m/s
- n = Refractive index of the medium

Step 2: Standard Test Case Table:

Test Case	Input(s) (Refractive Index, n)	Expected Output(s)	Actual Output(s)	Result (Pass/Fail)
1	1.00	300000000.00 m/s	300000000.00 m/s	PASS
2	1.33	225563909.77 m/s	225563909.77 m/s	PASS
3	1.50	200000000.00 m/s	200000000.00 m/s	PASS
4	2.00	150000000.00 m/s	150000000.00 m/s	PASS
5	2.42	123966942.15 m/s	123966942.15 m/s	PASS

Task 3: The Algorithm:

- Start.
- Declare two floating-point variables:
    - Refractive Index
    - speed
- Define the speed of light in vacuum as **300000000 m/s**.
- Read the refractive index from the user.
- Check whether the refractive index is greater than zero.
- If valid, calculate the speed of light using:
  Speed = Speed of Light in Vacuum ÷ Refractive Index
- Display the calculated speed.
- Otherwise, display an error message stating that the refractive index must be greater than zero.
- Stop.

Task 4: C program:

```
#include <stdio.h>

int main()
{
   const double SPEED_OF_LIGHT = 300000000.0;
   double refractive Index, speed;

   printf("Enter the refractive index: ");
   scanf("%lf", &refractive Index);

   if (refractive Index <= 0)
   {
      printf("Error: Refractive index must be greater than zero.\n");
      return 1;
   }

   speed = SPEED_OF_LIGHT / refractive Index;

   printf("\nSpeed of light in the medium = %.2lf m/s\n", speed);

   return 0;
}
```

Task 5: Test cases output:

Test case1:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter the refractive index: 1

Speed of light in the medium = 300000000.00 m/s
```

Test case2:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter the refractive index: 1.33

Speed of light in the medium = 225563909.77 m/s
```

Test case3:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter the refractive index: 1.5

Speed of light in the medium = 200000000.00 m/s
```

Test case4:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter the refractive index: 2

Speed of light in the medium = 150000000.00 m/s
```

Test case5:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter the refractive index: 2.42

Speed of light in the medium = 123966942.15 m/s
```